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Paleomagnetic and rock magnetic evidence for inverse zoning in the Parguaza batholith (southwestern Venezuela) and its implications about tectonics of the Guyana shield

Oscar E. Mirón Valdespino [1], Vincenzo Costanzo Alvarez

Departmento Ciencias de la Tierra, Universidad Simón Bolivar, A.P. 89000, Caracas 1060-A, Venezuela

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Abstract

We report paleomagnetic and rock magnetic data from the rapakivi granites of the Parguaza batholith (Guyana Precambrian Shield, southwestern Venezuela). These results suggest that the pluton is inversely zoned with respect to the cooling ages. In order to explain such an age pattern, tentative structural settings are proposed placing the Parguaza intrusions in a plate tectonic context. Six sites were sampled along a 200 km transect that cuts through the northern lobe of the batholith. Thermomagnetic curves, X-ray diffraction and fluorescence, hysteresis loops, thermal and alternating field (AF) intensity plots, transmission (TEM) and scanning (SEM) electron microscope analyses and Königsberger ratios (Q_n values) were used to identify the different magnetic mineralogies and their distribution of grain sizes. Magnetite, titanomagnetite near magnetite in composition and deuteric hematite are the three carriers of natural remanent magnetizations (NRMs) in these rocks. Magnetic granulometry indicators such as Königsberger ratios (Q_n values) suggest the dominant presence of single domain magnetites with average grain sizes grading from finer to coarser away from the center of the transect. The paleomagnetic results reveal the existence of two primary thermoremanent and/or thermochemical magnetizations (TRMs/TCRMs) for sites CSP-3, PI-2 and PI-4 (Decl.= 328°, Incl.= −21°, $k=15$, $\alpha_{95}=11.4°$) and sites CSP-2, PI-1 and PI-3 (Decl.=16°, Incl.=87°, $k=10$, $\alpha_{95}=13°$), respectively. There is also a poorly defined G3R$_2$ magnetization (Decl.=284°, Incl.= −86°, $k=6$, $\alpha_{95}=27°$) found in sites CSP-2, PI-1 and PI-3. The overlap of coercivities and the unblocking temperatures spectra, probably resulting from the coexistence of primary single-domain magnetic mineralogies with secondary exsolutions of single-domain-like Ti-poor (Fe-rich) regions (almost pure magnetite) in multidomain titanomagnetite grains, in most cases precludes the complete resolution of hybrid G1 (N+R) or G3 (N+R). The relative ages for these components were determined using a map of Rb/Sr model age 'chrontours'. G1 and G3 are the older and the younger TRMs/TCRMs, respectively, and were acquired at two discrete moments of the batholith's geological history. The final map resulting from integrating the paleomagnetic and Rb/Sr data, shows an age pattern for this batholith, and the rest of the intrusions that belong to the Parguaza Igneous Complex, that could be explained either as the effect of inverse cooling of a single intrusive body, the sequential emplacement of magmas controlled by normal faulting or an internal tectonism resulting in a system of NE-trending horsts and grabens cutting through the pluton. Because of its feasibility and agreement with the most recent theories about the tectonic evolution of the Guyana Shield, we favor the latter hypothesis. © 1997 Elsevier Science B.V.

[1]Present address: BP Venezuela, Edif. Seguros SudAmerica, El Rosal, Caracas, Venezuela.

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1. Introduction

The altitude dependence of magnetic remanence in slowly-cooled Precambrian terrains, caused by gradual uplift and erosion, has been extensively observed in some vertical sections in western Greenland and northwestern Scotland by Morgan (1976), Piper (1979, 1980, 1981) and Beckmann et al. (1977) among others. Similar studies have shown parallelism between thermal resetting of natural remanent magnetizations (NRMs) and K/Ar ages, thus developing a complete regional cooling history for part of the Grenville Province in southern Canada (e.g. Buchan et al., 1977; Berger et al., 1979). Most recently, Costanzo-Alvarez and Dunlop (1988) have described the cooling history of the Kapuskasing structural zone in northwestern Ontario (an uplifted cross-section through the upper and middle Archean crust) by combining paleomagnetism with $^{40}Ar/^{39}Ar$ thermochronometry. By the same token, namely by looking at the spatial distribution of time-calibrated paleomagnetic data, it should also be possible to establish the pattern of cooling for a batholith of considerable proportions. A continuous variation of distinctive vector directions for different components of remanence would be expected on a transect that cuts through the intrusion's internal zoning.

The purpose of this study is to describe the geological history and tectonic setting of the rapakivi granites comprising the Parguaza batholith, a large Precambrian intrusive body located in southwestern Venezuela (Fig. 1). The Parguaza batholith was chosen because it is one of the most thoroughly dated intrusive units in the Venezuelan Guyana Shield (Hurley et al., 1977; Olszewski et al., 1977; Gaudette et al., 1978; Barrios and Rivas, 1980; Barrios, 1983; Barrios et al., 1985; Gaudette and Olszewski, 1985). In addition, the geological and petrographic evidence for the metamorphism and alteration of these granites is almost nonexistent (Gaudette et al., 1978; Mendoza et al., 1977; Bangerter, 1985) which, besides the fact that rapakivi textures are thought to result from extremely slow cooling, with differences of ages as great as 200 Ma within a single intrusion (Dempster et al., 1994), makes the Parguaza batholith an ideal location for applying paleomagnetism in studying the zoning pattern of these kinds of Precambrian plutons.

Our paleomagnetic results are complemented not only by the radiometric data available (Rb/Sr), but also by magnetic mineralogy and rock magnetic analyses, which give information about the kind of NRM mineral carriers [X-ray diffraction and fluorescence, transmission (TEM) and scanning electron microscope (SEM) analyses and thermomagnetic curves] and their grain sizes [hysteresis parameters, alternating field (AF) and thermal demagnetization curves, Königsberger ratios (Q_n values), TEM and SEM analyses].

2. Geology and geochronology

The northern part of Amazonas state, in southwestern Venezuela, lies within the Guyana–Brazilian Shield of northern South America. This shield is comprised of various lithological units, has a complex petrotectonic character and has been subdivided into a number of geological provinces on the basis of lithological, metallogenic, tectonic and metamorphic similarities. The Parguaza batholith is located in the Cuchivero Province (middle Proterozoic), south of the Orinoco river. The Cuchivero Province comprises a high percentage of plutonic rocks and a few metavolcanic and metasedimentary units. One of its most distinctive features is the northwesterly strike that contrasts with the northeasterly structural trends of older regions such as the Archean Imataca complex (Gaudette et al., 1978). These trends are shown in Fig. 1 along with the location of the study area in the context of the whole southern Venezuelan Precambrian Shield.

The geological maps of Venezuela and South America (Bellizia et al., 1976; Wynn et al., 1993)

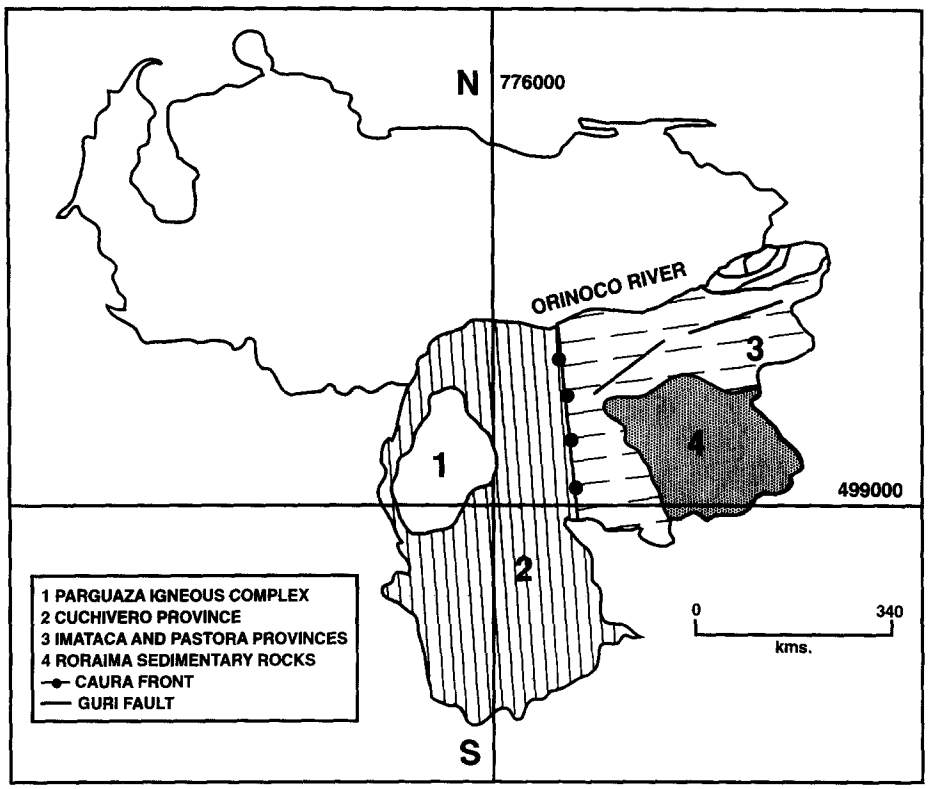


Fig. 1. Sketch map of Venezuela showing the tectonic setting of the PIC at the context of the whole Venezuelan Precambrian Shield (after Gaudette et al., 1978). It is noticeable the contrasting structural trends of the Cuchivero (NW) and the Imataca–Pastora geological Provinces (NE).

show, for the Cuchivero Province, a large extension of granitic intrusions known as the undifferentiated 'Parguazan granites' (referred to here as the 'Parguaza Igneous Complex'; PIC). From the geochronological evidence (Hurley et al., 1977; Olszewski et al., 1977; Gaudette et al., 1978; Barrios and Rivas, 1980; Barrios, 1983; Barrios et al., 1985; Gaudette and Olszewski, 1985), the intrusions of the PIC are the youngest of the Cuchivero Province. Thus, it seems quite reasonable to believe that they were not intruded by bodies of great magnitude which would result in regional reheating and consequent remagnetizations.

Mendoza et al. (1977) mapped the geological units that crop out in the northern part of the Amazonas state between the longitudes 616 000 to 813 000 m (UTM coordinates) and the latitudes 659 000 to 449 000 m. Beyond 659 000 m there is very little geological information available for the PIC (Bangerter, 1985; Menéndez et al., 1985). Fig. 2 (after Mendoza et al., 1977) shows some subdivisions for these undifferentiated intrusions together with other units such as the Roraima Group and the Santa Rosalia granite. The Roraima Group are sandstones dating from the Middle Proterozoic (Castillo and Costanzo-Alvarez, 1993). On the other hand, the Santa Rosalia-like intrusions have been considered as part of the igneous basement, intruded by the PIC, because of their widespread occurrence in the Guyana Shield, and evidence for metamorphism in these rocks (gray to pinkish highly foliated granites of variable granular textures).

Based on cross-cutting of mapped geological units (Fig. 2), it is possible to determine the relative ages of emplacement for the different plutons. From the oldest to the youngest are the Santa

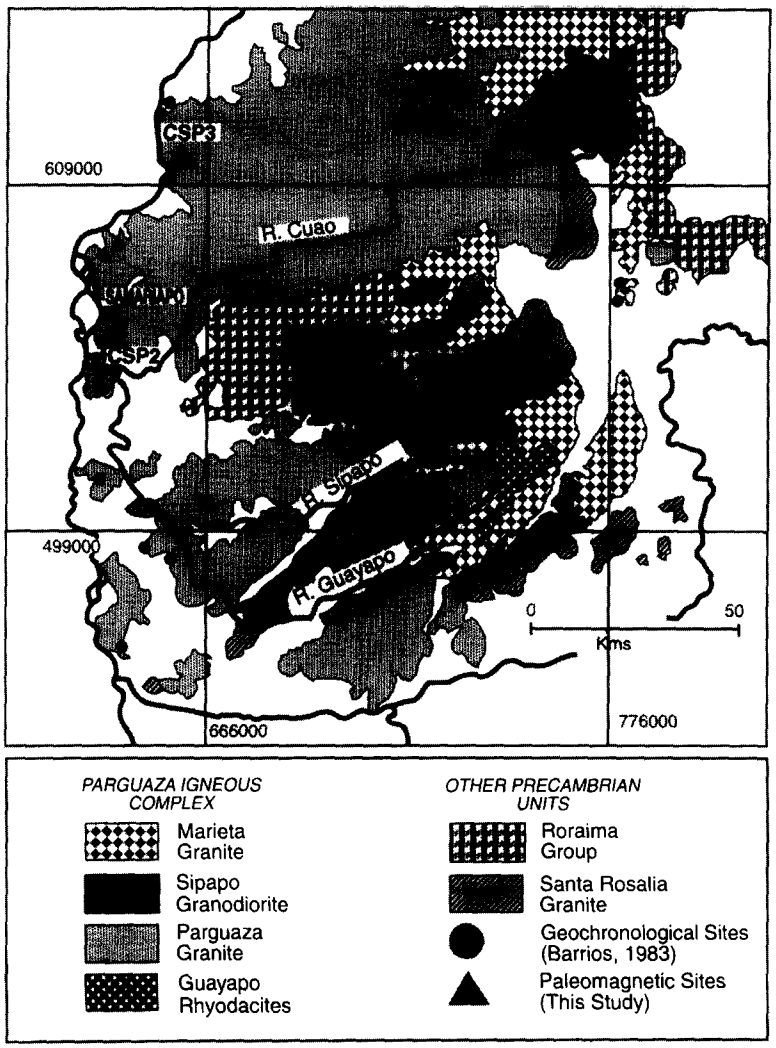


Fig. 2. Geological map of the southwestern area of study: PIC and other Precambrian units, after Mendoza et al. (1977) showing major geological units, paleomagnetic sites CSP-2 and CSP-3 (▲, this study) and geochronological sites (●, after Barrios, 1983).

Rosalia granite and the Parguaza batholith, the Sipapo granodiorite and the Marieta granite. It is not possible to determine if the Santa Rosalia granite is older or younger than the Parguaza batholith, nor the position of the Guayapo rhyodacites in the sequence of relative ages.

Using mineralogical, geochemical and visual analysis of Parguaza samples (recognition of some characteristic minerals and textures in the hand samples), Mendoza (1976) suggested that the rapakivi granites of the Parguaza batholith, the Sipapo granodiorites and the Guayapo rhyodacites represented a plutonic–hipoabysal–volcanic comagmatic sequence. If this hypothesis is correct it should be expected that the topography would reflect the petrology. In other words, the volcanic unit should be located at a higher paleotopographic level than the granodiorite, and the granodiorite, in turn, should lie above the Parguaza batholith. Although Fig. 2 does not show the topography it is noticeable that the Sipapo and Guayapo Rivers have their sources in the Sipapo granodiorite and the Guayapo rhyodacites, both at topographic levels higher than

those for the Parguaza rapakivi granites (Parguaza batholith).

The most extensive geochronological study of the PIC is that by Barrios (1983), who besides his own analyses, also shows a comprehensive compilation of the geochronological data available for the study area (Olszewski et al., 1977; Gaudette et al., 1978; Barrios and Rivas, 1980). As a matter of fact, some studies presented thereafter (e.g. Barrios et al., 1985; Gaudette and Olszewski, 1985) are mostly reinterpretations of these same data. According to the common practice of Hurley et al. (1977), Olszewski et al. (1977) and Gaudette et al. (1978), he assigns Rb/Sr model ages to a total of 19 sites distributed across the whole intrusive complex, assuming a comagmatic origin for the PIC and a $^{87}Sr/^{86}Sr$ initial ratio of 0.7050. Although they cannot be considered as absolute dates, model ages have been previously used in order to determine relative times of differential cooling in some Precambrian terrains such as the various lithotectonic domains of the Central Gneiss Belt in the Grenville Province of Canada (e.g. Dickin and McNutt, 1989, 1991). In the case of the PIC, some information could be drawn concerning the age-relationships of differentiation and the cooling for these intrusions. Fig. 3 shows a map of 'chrontours', a term coined by Piper (1980) to designate equal-age contours. This map has been generated using the Rb/Sr model ages available (after Barrios, 1983). The peaks represent the rocks that cooled first, whereas the valleys represent those which cooled last.

Barrios (1983) identifies as Guayapo rhyodacites the two geochronological sites located at the margins of the Guayapo River, and as Marieta granites the four easternmost geochronological sites shown in Figs. 2 and 3, all of them bearing relatively young model ages. However, according to the map of Fig. 2 (Mendoza et al., 1977), most of these sites should lie in the oldest Santa Rosalia granite. In order to settle this discrepancy, it is important to bear in mind that the evidence used in constructing the high-resolution geological map of Fig. 2 is mostly qualitative. In any case, providing that the Santa Rosalia granite is in fact an older igneous unit intruded by the PIC, we can still argue that the apparently anomalous young dates reported by Barrios for these sites, could be either initial cooling ages (if the sites were sampled indeed in the Guayapo rhyodacites and the Marieta granite) or metamorphic ages (if the sites were sampled in the Sta. Rosalia granite instead) resulting from the complete resetting of the isotope clocks at the time of emplacement of the younger PIC intrusions.

3. The Parguaza rapakivi granites and the paleomagnetic sampling

The Parguaza granites, comprised of the Parguaza batholith, are massive and coarse-grained rocks with an U/Pb zircons protolith age of 1545 ± 20 Ma (Gaudette et al., 1978). They represent the largest rapakivi intrusions in the world. About 10% of the phenocrystals of these rocks show multiple zoning, and in contrast with most rapakivi granites, the K-feldspar is microcline perthite instead of orthoclase. Epidote and chlorite are completely absent and the rocks show no visible signs of metamorphism subsequent to emplacement (Gaudette et al., 1978).

According to geochemical and petrogenetic studies, Mendoza (1974, 1976) suggested that the Parguaza batholith was formed by extensive fractional crystallization of a tholeiitic magma with a high alumina content. This magma probably came from the upper mantle and was mixed with charnockitic material from the lower crust. The granite melt rose up to epizonal levels close to the surface during the middle Proterozoic, under post-orogenic or epirogenic conditions (post-Transamazonic conditions) almost contemporaneously with the intrusion of the Roraima diabases (Gonzalez de Juana et al., 1980). According to Mendoza et al. (1977), the systematic increase of Mg/Fe ratios versus alkali percentage ($Na_2O + K_2O$), away from what seems to be the center of the batholith, suggests a pattern of inverse compositional zoning, grading from a mafic core that cooled first at temperatures close to 800°C, to felsic margins that cooled finally at temperatures of ca 700°C.

Based on the geochemical, petrogenetic and geochronological studies, Gaudette et al. (1978)

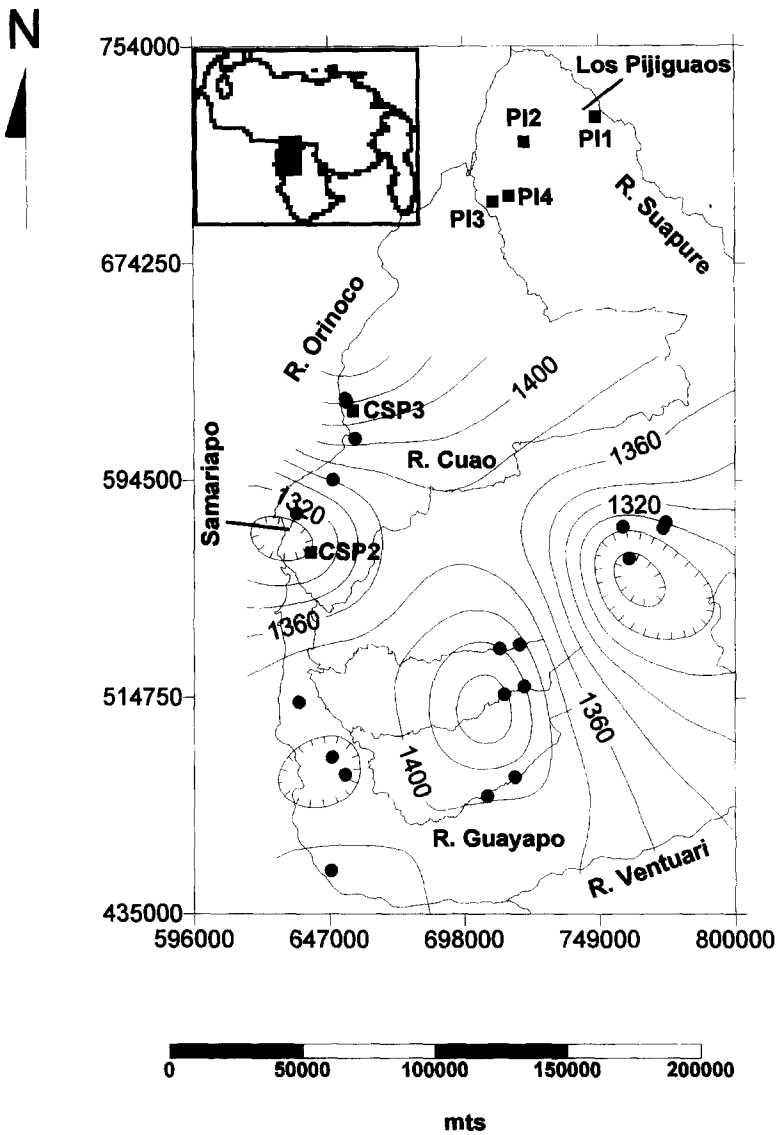


Fig. 3. "Chrontours" map [Rb/Sr model ages after Barrios (1983)] showing sampling sites (■) along a NE-striking transect that cuts through the northern lobe of the Parguaza batholith, and goes from the Los Pijiguaos area down to the Samariapo region. Geochronogical sites (●) in other intrusive bodies (after Barrios, 1983), are also shown.

hypothesized that the Parguaza batholith reflects within-plate extensional tectonics that prevailed during a period of ca 100 Ma in a large part of the northwestern and southern Guyana Shield (between 1500 and 1600 Ma). Rifting was probably accompanied by zones of high temperature gradient, slow migration of hot materials from subcrustal depths and deep faulting through the lithosphere, with the consequent growth of horsts and grabens. According to Gaudette et al. (1978) the Parguaza batholith was the result of anatexis by burial and volcanism-related heat flow from the sides and the base of a down-sinking block of trondhjemitic or charnockitic composition, developed in one of these extensional structures. Subsidence mechanisms of this kind have been

proposed as possible explanations for compositional inverse zoning in large granitic intrusions (Pitcher, 1993).

Most recently, Gaudette and Olszewski (1985) and Gibbs and Barron (1993) have proposed an alternative model to explain the origin of the Parguaza granites. They suggest that the emplacement of these rocks is the last phase of a 1500 Ma subduction that took place along a plate boundary whose actual suture (the Negro and Atabapo Rivers in southwestern Venezuela) is a long north–south belt of sediments metamorphosed in the lower greenschist facies. Thus, according to this model, the Parguaza granites would be considered as anorogenic intrusions in a subducted plate.

For the area north of 659 000 m of latitude, off the edge of Fig. 2, but south of site PI-3 (Fig. 3), Bangerter (1985) recognizes, as the main geological unit, an extension of the Parguaza batholith. She also points out that neither the field observations nor the petrographic analyses provide evidence for tectonic deformation. Also above 659 000 m of latitude, in the Los Pijiguaos region, north of site PI-3, the Los Pijiguaos granite crops out which is in nonreactional contact with the Parguaza batholith. For the Los Pijiguaos granite Menéndez et al. (1985) reported results for two drillings (DDH-1 and DDH-2) in which they reached an underlying rapakivi granite, probably Parguaza. This result suggests that Los Pijiguaos could be the roof of Parguaza. In this same area we sampled a syenite (PI-3) with a gradational contact towards a reddish granite that also shows a similarity with Parguaza.

In a preliminary paleomagnetic study of this batholith, Onstott (1981) reported three different directions of remanence: G1; G2; and G3. G1, an antipodal group with NW–SE declinations and shallow inclinations, was explained as a primary thermal remanent magnetization (TRM) acquired at the time of intrusion. G2, an antipodal group with NE–SW declinations and shallow to moderate positive and negative inclinations, was correlated with the Nickerian tecto-thermal event (ca 1250 Ma) on the basis of its similarity to secondary directions identified in the Imataca complex. G3, a poorly defined steep magnetization, with inclinations close to +90°, was interpreted as a secondary acquisition dating from the 'Braziliano' tectonic event because of the similarity of the corresponding pole position with the 500 Ma segment of McWilliams (1981) African polar wander path, after the appropriate rotation of South America to Africa to close the south Atlantic.

Our paleomagnetic sampling sites were located along the road that connects the southern port of Samariapo to the northeastern town of Los Pijiguaos (Fig. 3). The choice of such a transect was conditioned by the geological information available which suggests that all these sites lie on the Parguaza batholith. Also, the expedience of access ways to good outcrops guided our sampling. According to the hypothesis of Menéndez et al. (1985), the northeastern end of our sampling profile should be the roof of the batholith (Fig. 3). However, due to the little topographic contrast observed between Samariapo and Los Pijiguaos, it is difficult to estimate how deep into the pluton the transect goes as far as we move southwest.

A total of 46 oriented cores were collected at six sites (an average of seven samples per site) distributed along a distance of almost 200 km. Fig. 3 shows our sampling sites (filled squares), besides the Rb/Sr 'chrontours' and geochronological site locations (filled circles) (after Barrios, 1983). Only the southwestern section of the transect (sites CSP-2 and 3) crosses a documented age gradient. The rest of the sites, located on its northeastern end (PI-1, 2, 3 and 4), lie in a region for which there are no radiometric data available.

4. Magnetic characterization of the Parguaza batholith

X-ray diffraction analyses were carried out on a representative sample from each of the sites analyzed. Although these studies are a good initial approach to the identification of magnetic phases such as magnetite and hematite, they proved to be unsuitable for identifying titanomagnetites. X-ray fluorescence analyses performed in samples from each of our Parguaza sites revealed, in decreasing order of concentration, the presence of iron, zirconium and rubidium as the main metals, followed by potassium, calcium, titanium, vanadium,

chrome, manganese, zinc, gallium, strontium, yttrium, molybdenum and lead.

4.1. Rock magnetic analyses

The thermomagnetic curves shown in Fig. 4a and b are typical of the sample suite. They were measured in air using a saturating field of 200 mT. In most cases there is evidence for two Curie temperatures (e.g. the sample from site CSP-3, Fig. 4a). The lower Curie temperature (ca 400°C) is only seen during heating and reveals the presence of titanomagnetite. The second one appears in both heating and cooling curves at ca 580°C, revealing the presence of pure magnetite. Sites CSP-2 and PI-4 show no apparent signs of titanomagnetite, which is probably present in low concentrations and completely overshadowed by the high content of pure magnetite. Curie points of ca 400°C indicate titanomagnetites with $x=0.3$ (O'Reilly, 1976). Thus, according to the thermomagnetic analyses, the main magnetic minerals in these samples are concentrated around the magnetite end member of the magnetite-ulvospinel solid solution series, namely between $Fe_{2.7}Ti_{0.3}O_4$ and Fe_3O_4. The absence of titanomagnetite Curie points in all the cooling curves is evidence for the oxyexsolution of this mineral to magnetite during heating.

The percentage of saturation magnetization at temperatures of ca 700°C, for sites CSP-3 and PI-1, is low and close to 10% of the initial value measured at room temperature. However, the rest of the sites show saturation magnetizations over 30% of their initial values even at temperatures as high as 700°C (e.g. sample PI-4, Fig. 4b). This type of behavior could be due to the presence of a considerable fraction of superparamagnetic magnetite, hematite and/or pure iron.

Fig. 5 shows two hysteresis loops for representative samples from sites CSP-3 and PI-4. Hysteresis parameters also suggest the existence of a fraction of superparamagnetic or multidomain magnetite in most of these samples, since all sites analyzed show ratios of $J_{rs}/J_s \ll 0.1$ together with the low average $H_c = 7.5\ \text{mT} \pm 0.83$ (Dunlop, 1980).

Fig. 6 shows the AF and temperature total intensity decay plots of some representative samples from each site. The AF demagnetization curves have a typical single-domain form, with an initial plateau, inflected shape and high median destructive fields (Dunlop, 1983). Thermal demagnetization curves also reveal an initial plateau followed by a rather progressive decay of intensities over a range of unblocking temperatures between ca 400 and 600°C. This could be due to the presence of both fine-grained titanomagnetite and magnetite, plus a minor amount of hematite for those few samples where unblocking temperatures go over 580°C. The hematite must be either primary or deuteric, since the petrographic analyses of Parguaza samples show no evidence for late alteration or metamorphism (e.g. Bangerter, 1985; Mendoza, 1974, 1976).

Königsberger ratios (Q_n values) for each sample were calculated by measuring intensities of NRM and induced magnetizations (k_oH, for $H=$ 0.032 mT). Fig. 7 shows the site-average Q_n values plotted against the site location on the transect (site CSP2 has been taken as reference point). In weakly magnetized rocks, where the amount of magnetic minerals are a minimum percentage of the whole, Q_n values are not the ideal choice for determining relative grain sizes since magnetic susceptibility could be controlled and overshadowed by nonmagnetic phases as well (e.g. ilmenite, biotite and hornblende). However, that is not the case for the strongly magnetized Parguaza rocks.

Fig. 7 also shows the standard deviation for the site-averages of the Q_n values (dashed line). It is important to point out that Königsberger ratios for individual samples have experimental errors $<10\%$ of their values. Thus, in spite of the large standard deviations of each site-average, we can assert that all these Q_ns indeed have values >1, which is indicative of the presence of a dominant population of fine-grained magnetite (Stacey, 1967).

4.2. Magnetic mineralogy

Rock magnetic studies suggest that the main magnetic phases in this granite are single-domain to superparamagnetic magnetite and/or titanomagnetite, plus a fraction of deuteric hematite. TEM analyses for a sample from site PI-3, reveal the

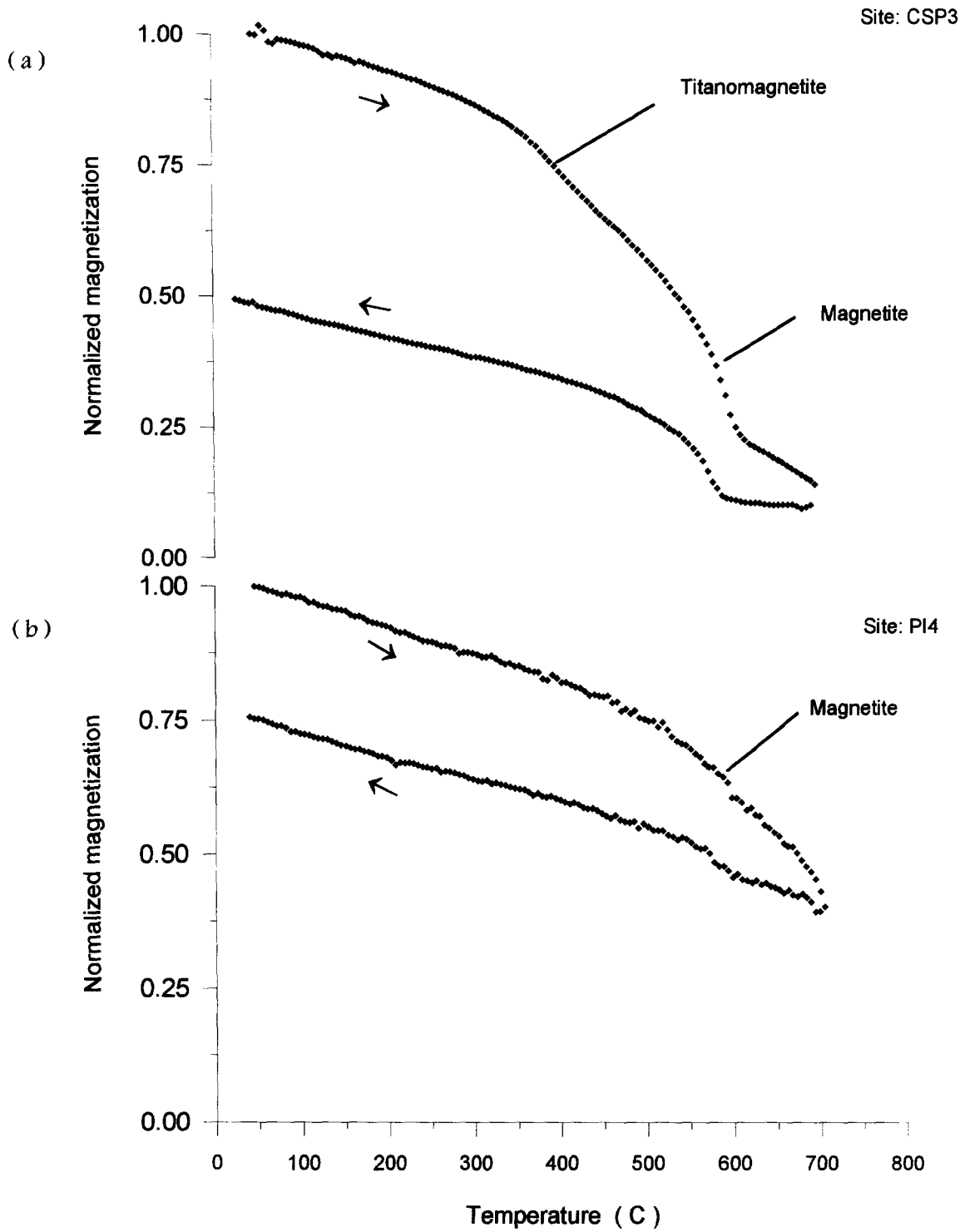


Fig. 4. Typical strong-field thermomagnetic curves for two representative samples from sites (a) CSP-3 and (b) PI-4.

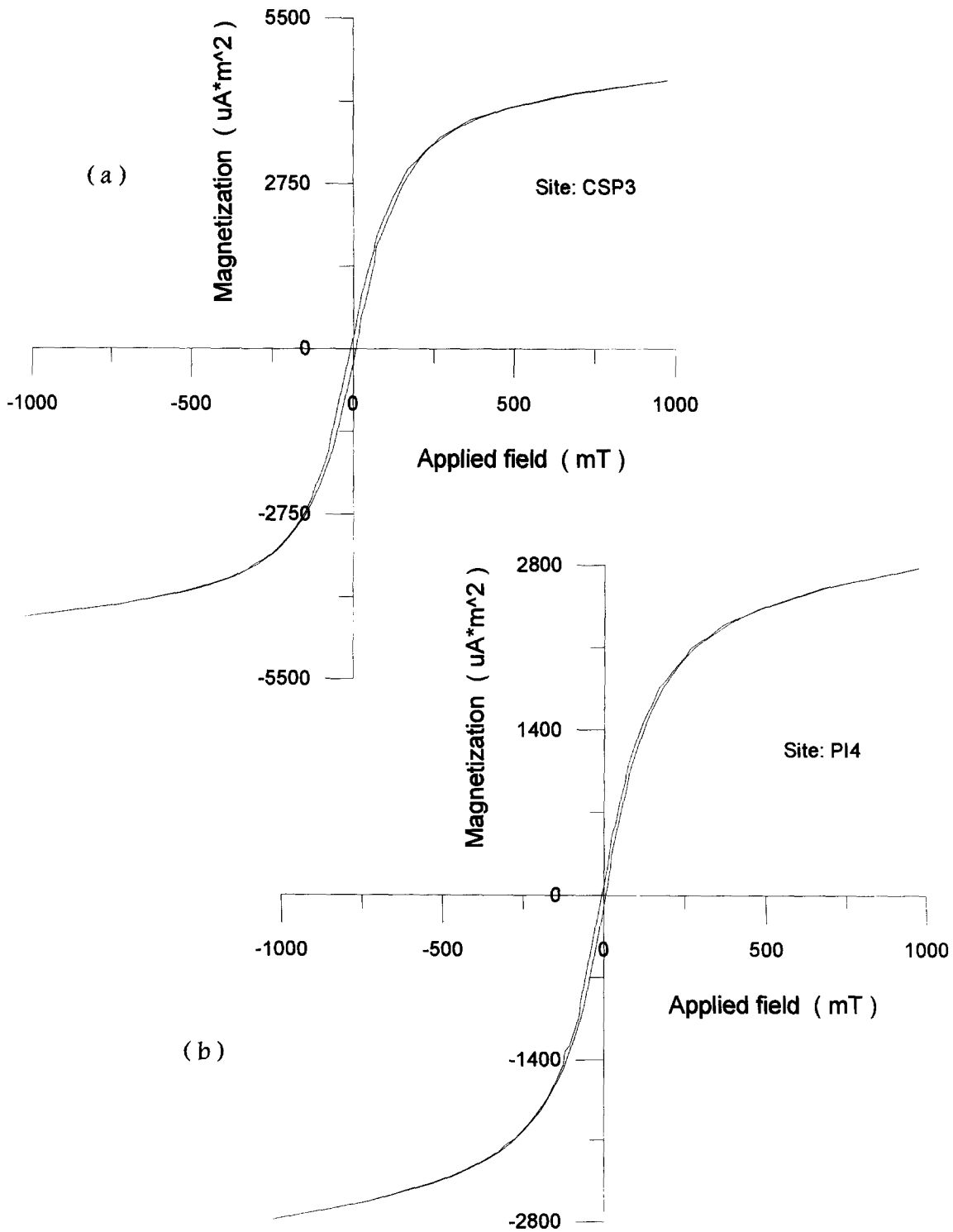


Fig. 5. Typical hysteresis curves for two representative samples from sites (a) CSP-3 and (b) PI-4.

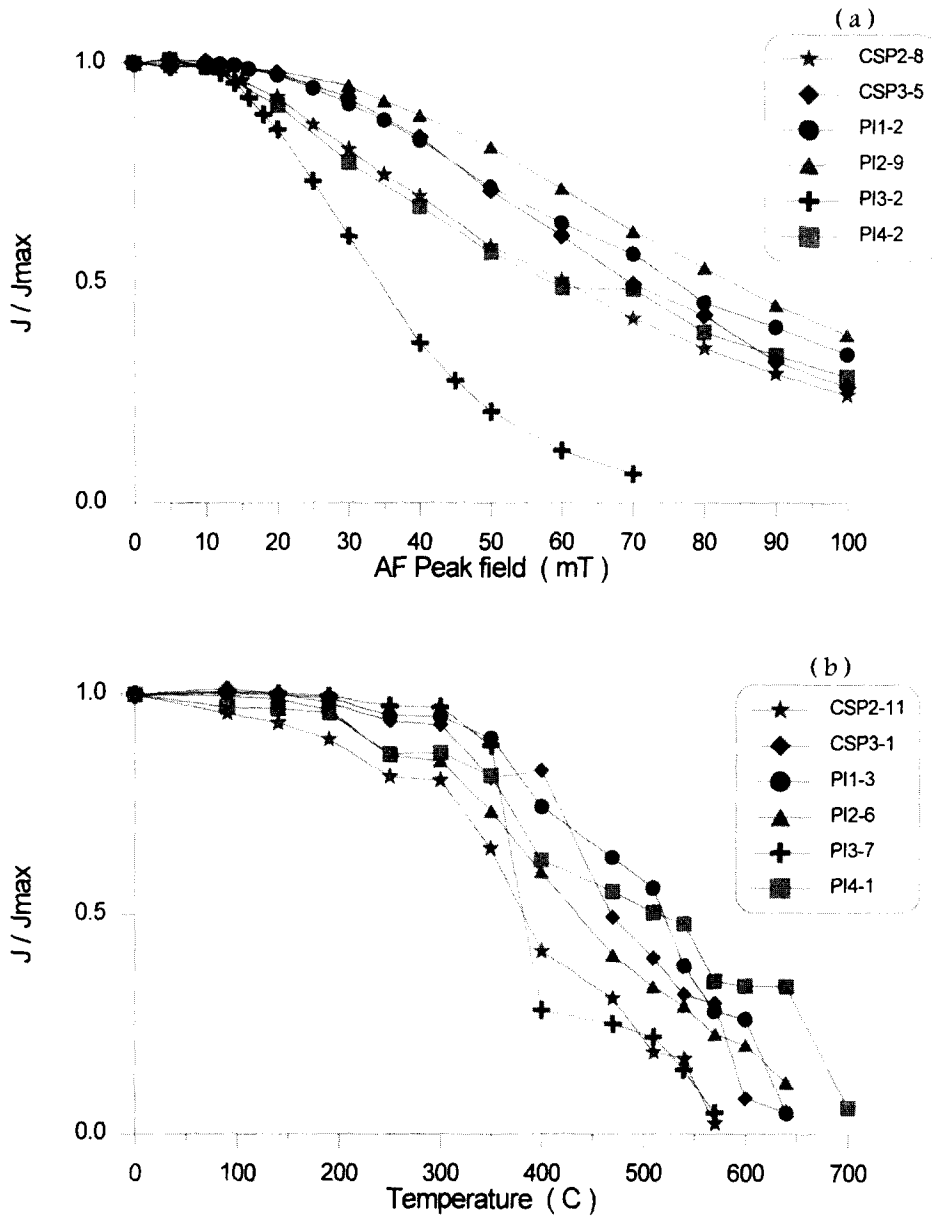


Fig. 6. (a) Typical coercivities and (b) unblocking temperatures spectra of a representative sample from each site analyzed.

presence of these Fe and Ti-rich fine-grained opaque minerals of submicron sizes.

On the other hand, the transmitted light petrography and visual analysis of the hand samples show a considerable amount of coarse-grained opaque minerals (magnetite and/or titanomagnetite). However, the expected response of multidomain magnetite or titanomagnetite in the AF and thermal demagnetization curves, hysteresis parameters and Q_n values, is almost absent from most samples.

Scanning electron microscope analyses were carried out in order to examine in detail some of these large opaque minerals. Fig. 8a is a SEM photomicrograph (topographic image of secondary electrons) of an ca 0.1 mm titanomagnetite

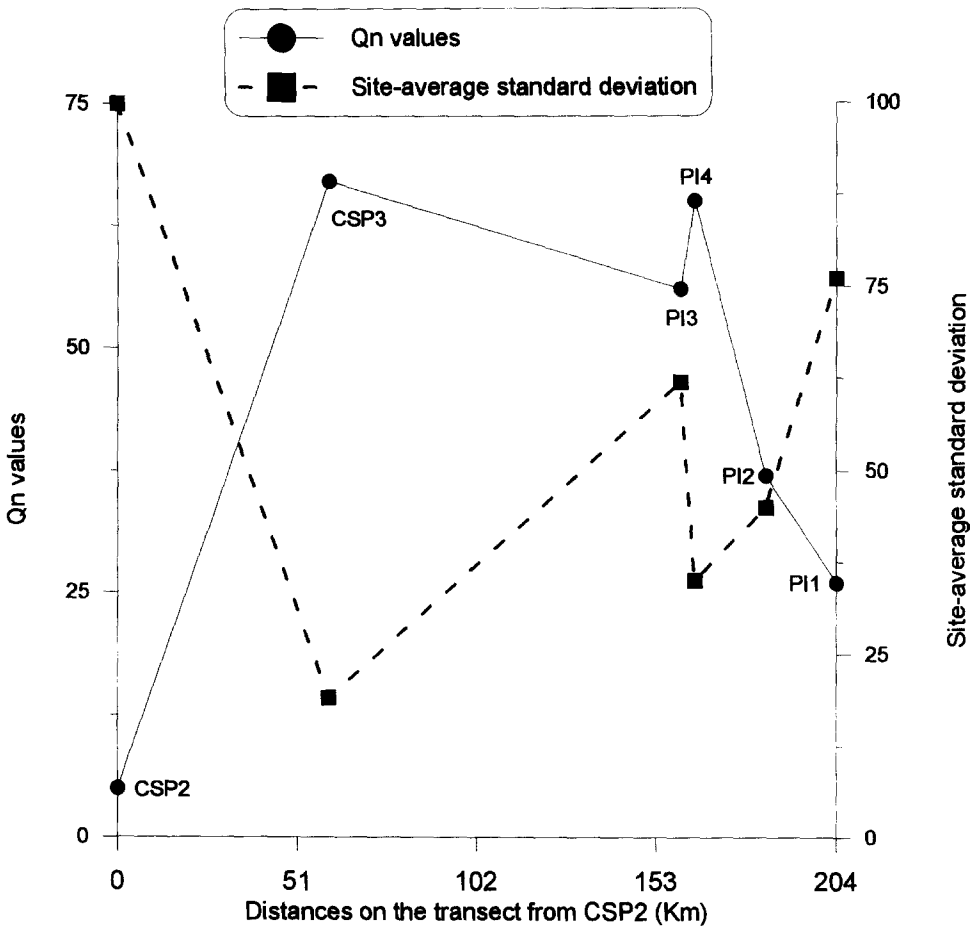


Fig. 7. Site-averaged Königsberger ratios (●) and the standard deviation of these averages (■) plotted against the position of these sites on the transect with respect to CSP2.

crystal from site PI-3. Lamellae of Ti-rich (Fe-poor) lens-like exsolution bodies of submicron sizes are seen on (111) planes of Ti-poor (Fe-rich) titanomagnetite crystals. Fig. 8b is a magnification of some of these Ti-rich intergrowths. A profile of Ti concentrations (X-ray energy dispersion spectra analyses) has been superimposed to the picture, showing variations that go from $Fe_{2.3}Ti_{0.7}O_4$ in the exsolution bodies themselves (darker stripes), down to almost pure magnetite ($Fe_{2.88}Ti_{0.12}O_4$) in between (lighter stripes). In other cases, the Ti-rich regions could be also ilmenite intergrowths produced by high-temperature (just above 580°C) oxidation of homogeneous titanomagnetite (El Goresy et al., 1976). Thus, the characteristic single-domain rock magnetic properties observed in most Parguaza samples must be the response not only to the presence of primary fine-grained magnetite, titanomagnetite and hematite, but also to the presence of a generation of multidomain titanomagnetites with secondary internal exsolutions of pure, or almost pure, magnetite of submicron sizes alternating with titanomagnetite intergrowths.

Mineral exsolutions are common features in granites. In fact, it is a well known fact that rapakivi granites cool at very slow rates and at relatively low temperatures. These cooling conditions allow the exsolution of albite and oligoclase in K-feldspars (Dempster et al., 1994). Also in granitic bodies, hematite lamellae in primary

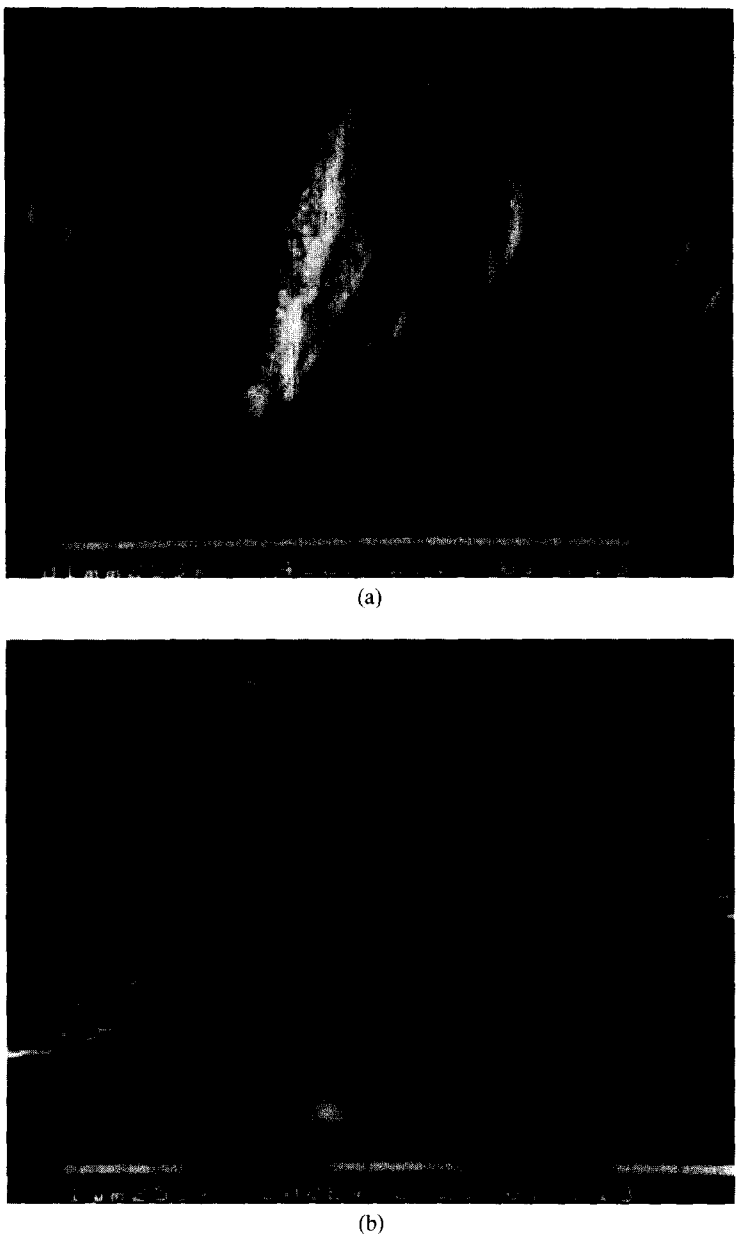


Fig. 8. (a) SEM photomicrograph of a titanomagnetite crystal from site PI-3. Lamellae of Ti-rich (Fe-poor) lens-like exsolution bodies of submicron sizes, form on (111) planes of Ti-poor (Fe-rich) titanomagnetite. (b) A magnification of some of these Ti-rich intergrowths. A profile of Ti concentrations (X-ray energy dispersion spectra analyses) has been superimposed to the picture showing variations that go from $Fe_{2.3}Ti_{0.7}O_4$ at the darker stripes, down to almost pure magnetite ($Fe_{2.88}Ti_{0.12}O_4$) at the lighter stripes.

magnetite grains or ilmenite exsolutions in hematite, have been widely observed (e.g. Dunlop et al., 1984; Costanzo-Alvarez and Dunlop, 1993; Benn et al., 1993).

5. Paleomagnetic results

Thermal and AF stepwise demagnetizations were carried out on selected specimens. AF demag-

netizations at 5 mT increments up to 100 mT were carried out on all samples. However, in those cases where NRM intensities remained high at 100 mT, further thermal demagnetization was applied up to 600°C. Standard Schonstedt AF demagnetizers and furnaces were used for AF and thermal demagnetizations. NRM measurements were performed in a Molspin spinner magnetometer.

Figs. 9 and 10 show the typical demagnetization response of high coercivities and unblocking temperature NRMs in representative samples from sites CSP-3, PI-4, CSP-2 and PI-3. With the exception of CSP-3-5 (Fig. 9a), which has a soft magnetic overprint of viscous origin, the rest of these samples exhibit univectorial NRMs with slow AF decay (median destructive fields ranging from 15 to 100 mT) and discrete unblocking temperatures restricted to narrow intervals between ca 400 and 600°C. According to these curves, the NRM mineral carriers should be single-domain magnetite and/or titanomagnetite in a narrow range of grain sizes.

Fig. 11 shows the typical AF demagnetization response in bi-vectorial samples CSP-3-4 and PI-3-3. In both cases a softer magnetization with coercivities <30 mT is removed to reveal the underlying harder remanence.

The rather smooth transition between the two magnetizations carried by sample CSP-3-4 (Fig. 11a) suggests a thermochemical origin for

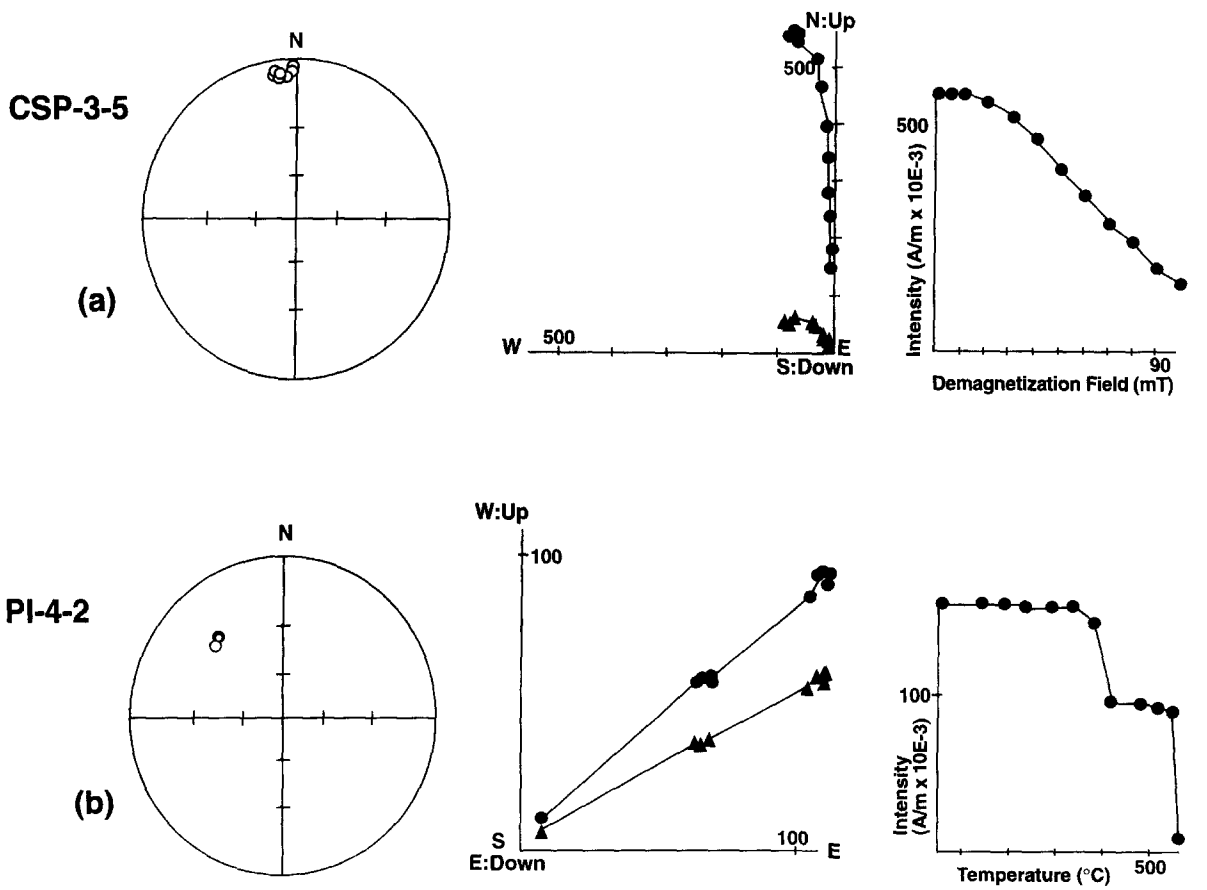


Fig. 9. Orthogonal vector and equal area projections of magnetization vectors in the course of AF and thermal demagnetization, and the corresponding total intensity decay curves, are shown for two representative specimens of sites (a) CSP-3 and (b) PI-4. In the orthogonal vector projections, circles are horizontal plane projections and triangles are vertical plane projections.

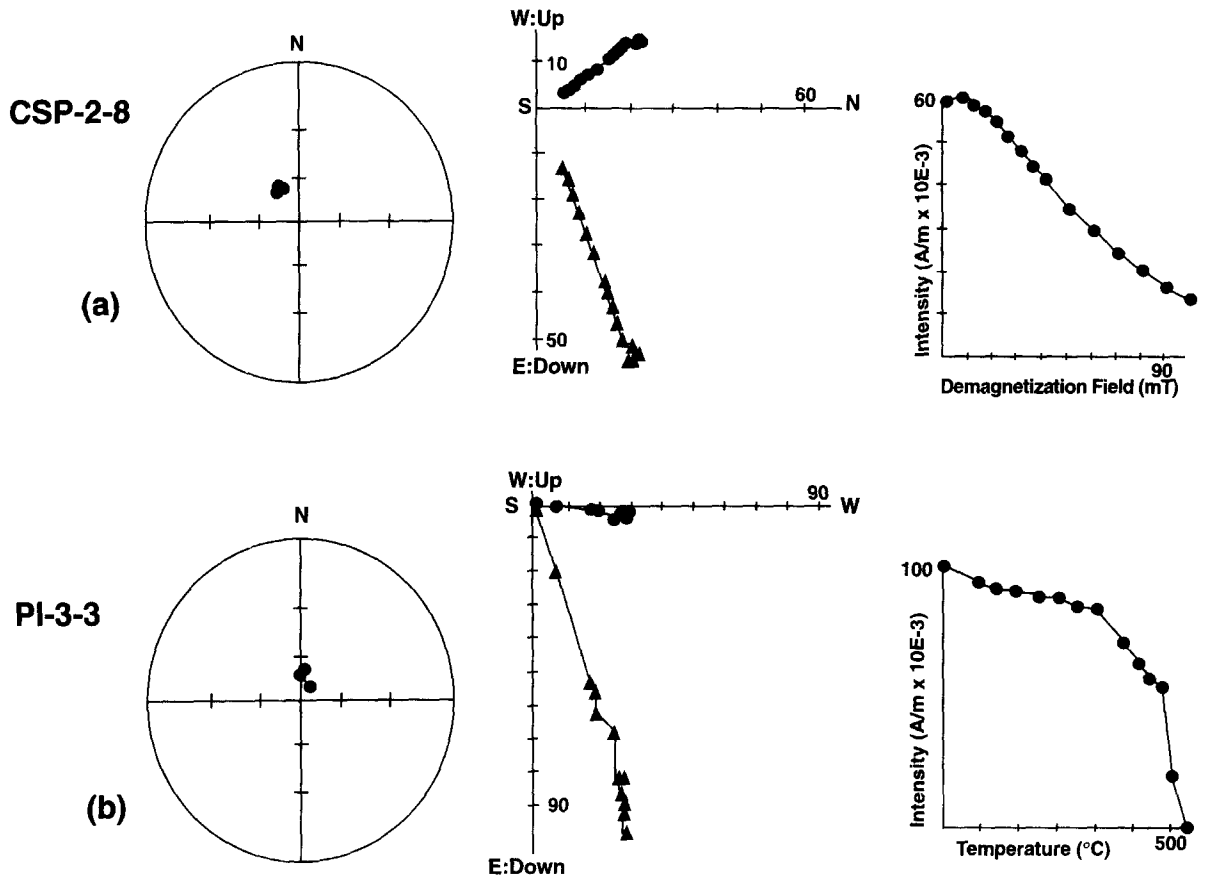


Fig. 10. Orthogonal vector and equal area projections of magnetization vectors in the course of AF and thermal demagnetization, and the corresponding total intensity decay curves, are shown for two representative specimens of sites (a) CSP-2 and (b) PI-3. In the orthogonal vector projections, circles are horizontal plane projections and triangles are vertical plane projections.

the soft overprint (McClelland-Brown, 1982). On the other hand, the Zijderveld plot for sample PI-3-3 (Fig. 11b) indicates that these two superimposed NRMs have almost antiparallel directions. This is also reflected in the dramatic increase of total intensity up to 30 mT upon AF stepwise demagnetization.

Approximately 50% of the Parguaza samples show streaking along great circle trajectories and never reach stable end points (e.g. the great circle paths shown in Figs. 12 and 13). Another 34% of these samples reach stable end points (e.g. filled diamonds in Figs. 12 and 13) with recognizable G1 or G3-like magnetizations (Onstott, 1981). There is also 16% of these samples that reach stable end points with nonrecognizable normal directions of magnetization, lying, in some cases, on the remagnetization circle of another sample (e.g. filled squares in Fig. 12).

Stable end points translate into linear segments that go through or near the origin of a Zijderveld plot (Zijderveld, 1967). Kirschvink principal component analyses (Kirschvink, 1980) were performed for these high coercivities and unblocking temperature magnetizations in order to obtain a sample mean direction (DirKir in Tables 1 and 2).

Those samples that reach stable end points with recognizable G1-like directions of remanence come exclusively from sites CSP-3, PI-2 and PI-4, and those with G3-like stable end points from sites CSP-2, PI-1 and PI-3. Final G1R and G3N mean directions for these two groups of sites (Table 3,

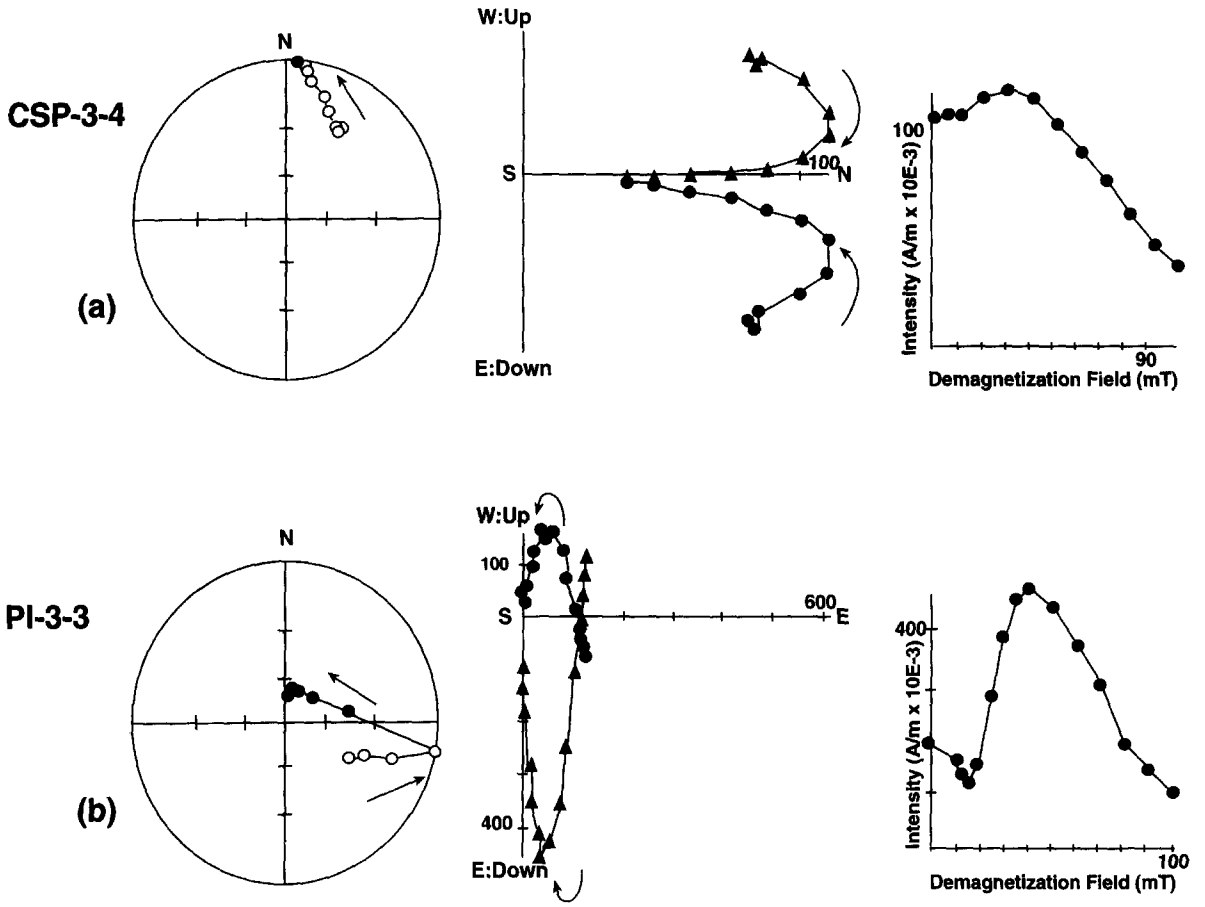


Fig. 11. Orthogonal vector and equal area projections of magnetization vectors in the course of AF demagnetization and the corresponding total intensity decay curves, are shown for two representative specimens of sites (a) CSP-3 and (b) PI-3. In the orthogonal vector projections, circles are horizontal plane projections and triangles are vertical plane projections.

Fig. 14), are calculated using a combined analysis that includes the Kirschvink mean directions for samples with recognizable stable end points, and the intersection of remagnetization circles for samples whose NRMs do not reach stable end points (McFadden and McElhinny, 1988). Differences of magnetic susceptibility anisotropy between samples, as well as secular variations of the earth's magnetic field, could account for the within-group dispersion observed in the final means (Fig. 14).

Great circle trajectories (GCnKir in Tables 1 and 2) from Figs. 12 and 13a show only partial G1N→G1R and G3R→G3N reversals upon AF or thermal stepwise demagnetizations. G1R and G3N-like intersections of remagnetization circles in these figures are anchored by the Kirschvink directions of those samples with recognizable stable end points. However, that is not the case for their corresponding later overprints G1N and G3R, which are completely unconstrained by the paleomagnetic data available. There is also a poorly defined direction of remanence we call $G3R_2$ (Fig. 13b) that seems to be either a recording of an incomplete reversal of the geomagnetic field that occurred immediately before the acquisition of G3N, or a G3R NRM residing in magnetic mineralogies of higher coercivities and unblocking temperatures than those carrying G3N magnetizations.

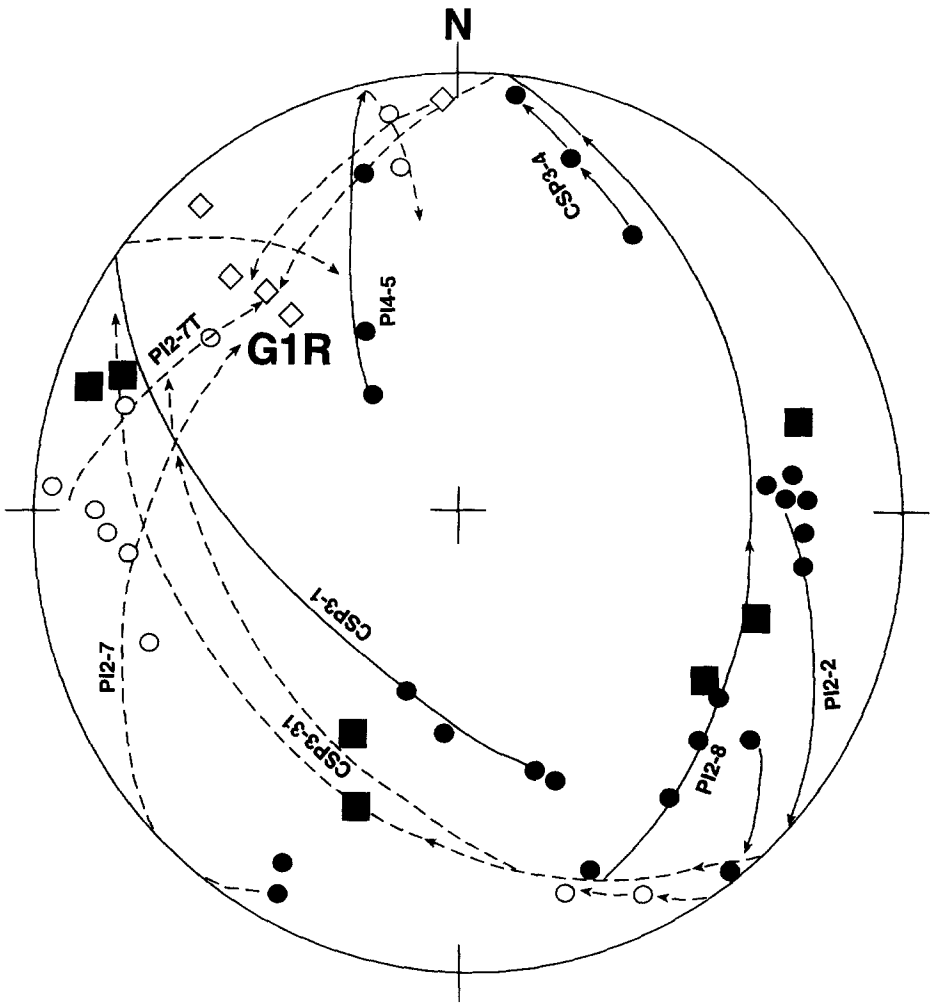


Fig. 12. Equal area projection of typical demagnetization trajectories of NRM vectors in the course of AF or thermal cleaning for those Parguaza samples that do not reach a stable end point and whose demagnetization trajectories seem to converge towards a G1R-like direction of remanence (sites CSP-3, PI-2 and PI-4). Kirschvink directions for samples that reach a G1R-like end point (♦) and a non-recognizable end point (■) are also shown.

6. Discussion

6.1. G1 and G3 magnetizations: inferred mechanism of TRMs/TCRMs acquisitions

Those samples that show streaking along a great circle path in the course of AF and thermal demagnetizations, or those whose stable end points are nonrecognizable directions of remanence, can be either TRMs recorded at different stages of a polarity transition of the geomagnetic field, or just unresolved hybrid magnetizations lying between a normal paleomagnetic component and its reversed counterpart. The first hypothesis requires fast cooling intrusions which is not the case for rapakivi granites. We favor the second hypothesis instead, which requires a complete overlap of coercivities and unblocking temperature spectra for the two superimposed directions of remanence. In the particular case of those samples showing nonrecognizable stable end points the situation is even more restrictive since the two spectra of coercivities or unblocking temperatures must be also entirely coincident. Onstott's G2 magnetization, an

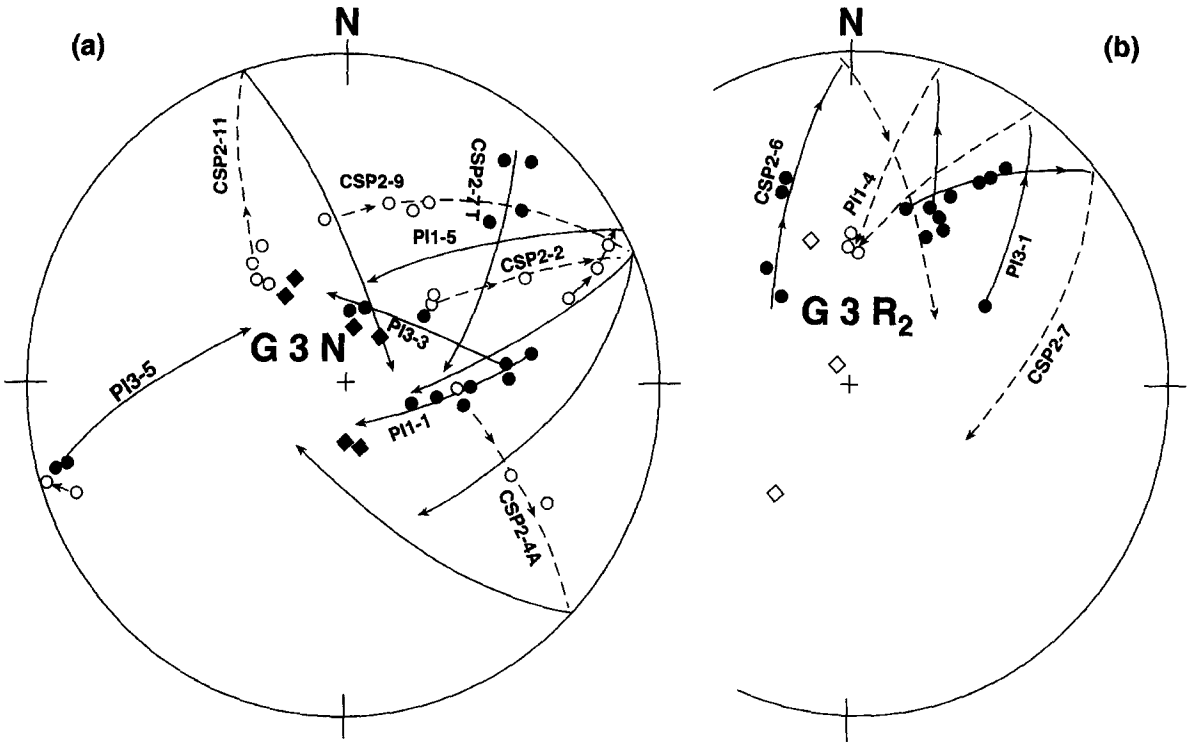


Fig. 13. Equal-area projection of typical demagnetization trajectories of NRM vectors in the course of AF or thermal cleaning for those Parguaza samples that do not reach a stable end point and whose demagnetization trajectories seem to converge towards a (a) G3N-like or (b) $G3R_2$ direction of remanence (sites CSP-2, PI-1 and PI-3). Kirschvink directions for samples that reach a G3N-like end point (◆) are also shown.

antipodal NE NRM, seems to be one of these stable hybrid components lying in the midway between normal and reversed G3 directions.

In order to explain the overlaps of demagnetization spectra in our Parguaza samples (ca 66% of the collection), we suggest the coexistence of two generations of magnetic grains of similar or equal shape and/or size distributions. One of them carries either G1R or G3N magnetizations, and the other one carries their antipodal counterparts. The G1R or G3N carriers would be the primary magnetic mineralogies identified in these samples, namely fine-grained magnetite or titanomagnetite plus a minor and variable fraction of deuteric hematite. The secondary magnetic phase (carrying G1N or G3R) is probably the one produced, not long after the acquisition of primary remanences, by the single-domain-like alternating pattern of almost pure magnetite and titanomagnetite exsolution bodies within primary multidomain titanomagnetites (Fig. 8).

Titanomagnetites unmix at relatively low temperatures $<580°C$ and the exsolution of Ti-rich and Ti-poor regions in magnetic grains by the diffusion of Fe and Ti cations is only observed in slowly cooled plutonic rocks (Butler, 1992). A slow cooling rate also allows deuteric oxidation to reach completion in the 1000–580°C range (Carmichael, 1982). Thus, since 580°C is not only the temperature at which secondary exsolutions and oxidation take place, but is also the maximum temperature at which TRMs are acquired by primary single-domain magnetites, we should expect either a unique or two almost-synchronous magnetizations residing in these two generations of magnetic minerals. As a matter of fact, polarity transitions of the geomagnetic field such as G1N→G1R and G3R→G3N, are usually quick

Table 1
Component G1 remagnetization circles and Kirschvink directions

Sample	Analysis	Treatment	Decl. (°)	Incl. (°)	ErAng (°)
CSP3-1T	GCnKir	Thermal	39.5	30.8	5.5
CSP3-3-1	GCnKir	AF	234.7	42.6	7.8
CSP3-5	Dir Kir	50–100 mT	357.7	−5.3	1.0
CSP3-4	GCnKir	AF	275	45.5	3.5
CSP3-6	Dir Kir	50–100 mT	319.8	−5.5	1.8
PI2-2	GCnKir	AF	225.2	57.9	30.8
PI2-7	GCnKir	AF	314.4	56.1	13.9
PI2-7T	GCnKir	Thermal	349.5	54.7	16.0
PI2-8	GCnKir	AF	275.8	55.7	9.6
PI4-1A	Dir Kir	60–100 mT	320.9	−28.8	3.7
PI4-1BT	Dir Kir	470–700°C	321.0	−35.4	4.2
PI4-2T	Dir Kir	510–570°C	317.8	−23.0	0.4
PI4-5	GCnKir	AF	78.6	13.0	2.3
Hybrid stable end points					
PI2-3T	Dir Kir	350–570°C	204.1	38.4	2.2
PI2-4T	Dir Kir	400–570°C	107.6	31.6	4.1
PI2-6T	Dir Kir	510–640°C	212.0	46.2	3.5
PI2-8T	Dir Kir	350–570°C	122.7	36.5	3.7
PI2-9A	Dir Kir	70–100°C	294.0	15.1	3.2
PI2-9BT	Dir Kir	510–570°C	292.4	10.1	2.4
PI4-4T	Dir Kir	510–570°C	73.7	22.4	0.9

Note: GCnKir are great circle trajectories fitted to the demagnetization data. Declination and Inclination are the coordinates of their poles.
DirKir are Kirschvink directions determined by principal component analyses on high coercivity and unblocking temperature demagnetization data.

Table 2
Component G3 remagnetization circles and Kirschvink directions

Sample	Analysis	Treatment	Decl. (°)	Incl. (°)	ErAng (°)
G3N					
CSP2-2T	GCnKir	Thermal	335.3	10.4	9.7
CSP2-4	GCnKir	AF	39.9	17.7	17.2
CSP2-7T	GCnKir	Thermal	299.8	22.7	26.5
CSP2-8A	Dir Kir	50–100 mT	322.5	64.1	1.6
CSP2-8BT	Dir Kir	300–400°C	326.0	63.5	8.3
CSP2-9	GCnKir	AF	333.9	40.1	7.4
CSP2-11	TGCnKir	Thermal	248.1	13.1	23.8
PI1-1	GCnKir	AF	337.1	10.9	3.7
PI1-2	Dir Kir	70–100 mT	155.6	72.8	2.0
PI1-3T	Dir Kir	570–640°C	36.1	72.7	0.5
PI1-5	GCnKir	AF	148.3	18.5	5.8
PI3-3	GCnKir	AF	191.8	19.4	3.8
PI3-3T	Dir Kir	570–640°C	4.1	71.2	0.5
PI3-4T	Dir Kir	300–600°C	257.4	71.3	1.6
PI3-5	GCnKir	AF	157.4	21.5	9.0
G3R$_2$					
CSP2-3T	Dir Kir	400–530°C	320.3	−85.4	1.2
CSP2-6T	GCn Kir	Thermal	84.6	21.4	12.9
CSP2-7	GCn Kir	AF	132.1	27.4	9.3
PI1-4	GCn Kir	AF	283.3	13.2	12.9
PI3-1T	GCn Kir	Thermal	302.9	21.3	6.6
PI3-2	Dir Kir	45–70 mT	339.6	−51.8	1.3
PI3-6	Dir Kir	40–100 mT	214.4	−54.5	1.2

Note: GCnKir and DirKir as in Table 1.

Table 3
Final mean directions calculated by using McFadden and McElhinny (1988) combined analysis of remagnetization circles

Component	Decl. (°)	Incl. (°)	k_s	α_{95} (°)
G1	328	−21	15	11.4
G3N	16	87	10	13
G3R$_2$	284	−86	6	27

events occurring within a time-span of no more than 5000 years.

6.2. *Inferred magnetic and geological history of the Parguaza batholith and the PIC*

According to Mendoza et al. (1977) and Martín (1972) the only orogenic event in Venezuela during Precambrian times, occurred after the intrusion of the Parguaza rapakivi granites (Parguaza batholith), was the Nickerian event (ca 1250 Ma). Alternatively Olszewski et al. (1977) proposed, on the basis of U/Pb and Rb/Sr analyses of granites and gneisses from the southern margin of the batholith, that a 400–500 Ma thermal event also affected this region. However, there is no field or petrographic evidence for alteration or metamorphism in the Parguaza batholith (e.g. Gaudette et al., 1978; Mendoza et al., 1977; Bangerter, 1985). We argue that G1 and G3 (N and R) are both primary TRMs/TCRMs dating from the initial stages of cooling of the rapakivi granites comprised by this batholith.

Individual samples that undergo cooling at a sufficiently slow rate may record apparent polar wandering over their unblocking temperature spectra, with a consequent streaking in the

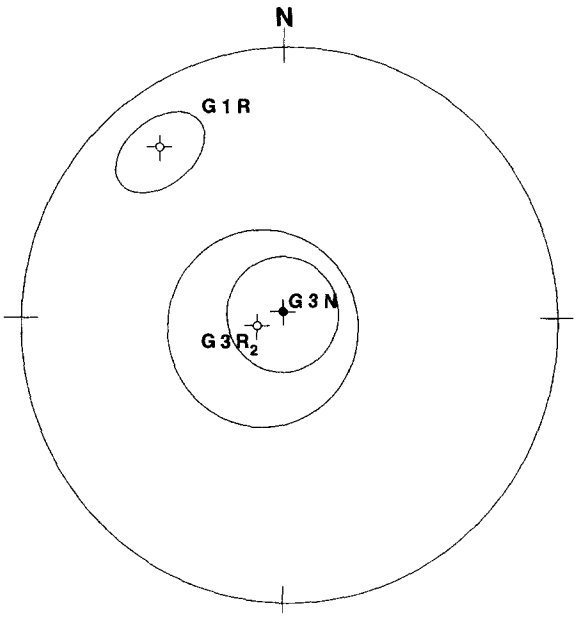


Fig. 14. Equal-area projections of McFadden means (McFadden and McElhinny, 1988) for G1(N and R), G3 (N and R) and $G3R_2$ calculated by using intersection of remagnetization circles and Kirschvink mean directions for samples that reach G1R, G3N or $G3R_2$-like stable end points.

directions of remanence between two apparently stable end points. However, such streaking only occurs in samples with a wide spectrum of unblocking temperatures due, for instance, to a continuous variation in grain size or ionic substitution of their magnetic minerals (Morgan, 1976). In the Parguaza samples the unblocking temperatures for G1 and G3 are restricted to narrow intervals between ca 400 and 600°C. This could explain why there is only a discrete record of two different moments in the geological history of the batholith, namely G1(N and R) and G3 (N and R).

The map of Rb/Sr 'chrontours' (Fig. 3) can be used to assign relative ages to G1 (N and R) and G3 (N and R). The only 'datable' sites are CSP-2 and CSP-3 since the rest of them lie in the Los Pijiguaos region where there is no radiometric data available. Site CSP-2 is in an age valley (ca 1300 Ma) whereas site CSP-3 lies close to the 1440 Ma 'chrontour'. Thus we can say that G1 (site CSP-3) and G3 (site CSP-2) reflect, respectively, an early and a later stage of cooling of the Parguaza rapakivi granites.

By assigning 1300 Ma to PI-1 and PI-3 (the other two sites, besides CSP-2, for which only a G1 TRM/TCRM has been isolated) and 1440 Ma to sites PI-2 and PI-4 which, together with CSP-3, make up the group of sites that shows solely a characteristic G3 TRM/TCRM, we can extrapolate the map of 'chrontours' up to the Los Pijiguaos region (Fig. 15) and thus obtain a complete picture of the sequence of cooling events of the Parguaza batholith.

The 'chrontours' maps of Figs. 3 and 15 were constructed applying the geostatistical method of ordinary kriging on the radiometric data available (Program SURF 5.0 by Golden Software Inc. 1994). We used a simple search procedure on a 15×10 grid with a separation between nodes of ca 22.6 km. In order to assign average dates to these nodes, only their nearest four data points were taken into account after trying out an inspection radius of 320 km.

This map shows an area between sites CSP-3 and PI-3 that apparently cooled first. On either side of it the cooling ages are younger. The pattern of relative ages observed in Fig. 15 indicates that the Los Pijiguaos region was the last to cool whereas the rest of the sites (with the exception of CSP-2) cooled first. Moreover, at a regional scale, a conspicuous zone of high-age values, flanked at both sides by two lower-age terrains, shows up clearly in the whole 'chrontours' map of Fig. 15. These high and low-age zones comprise, at their northern lobes, some of our sampling sites in the Parguaza batholith, and stretch down to include the southern geochronological sites in other intrusive bodies of the PIC (Barrios, 1983).

Alternatively, the general trend for the Q_n values observed in Fig. 7 seems to indicate that the average grain sizes of magnetite at the center of the transect are smaller than those at its ends. Assuming these grain sizes have not been significantly changed by later alteration, which is reasonable to believe in view of the petrological, rock magnetic and paleomagnetic evidence available, then this result reveals a textural zoning for the Parguaza rapakivi granites probably related to differential cooling.

Some interesting conclusions can be drawn by integrating the regional geology and the rock

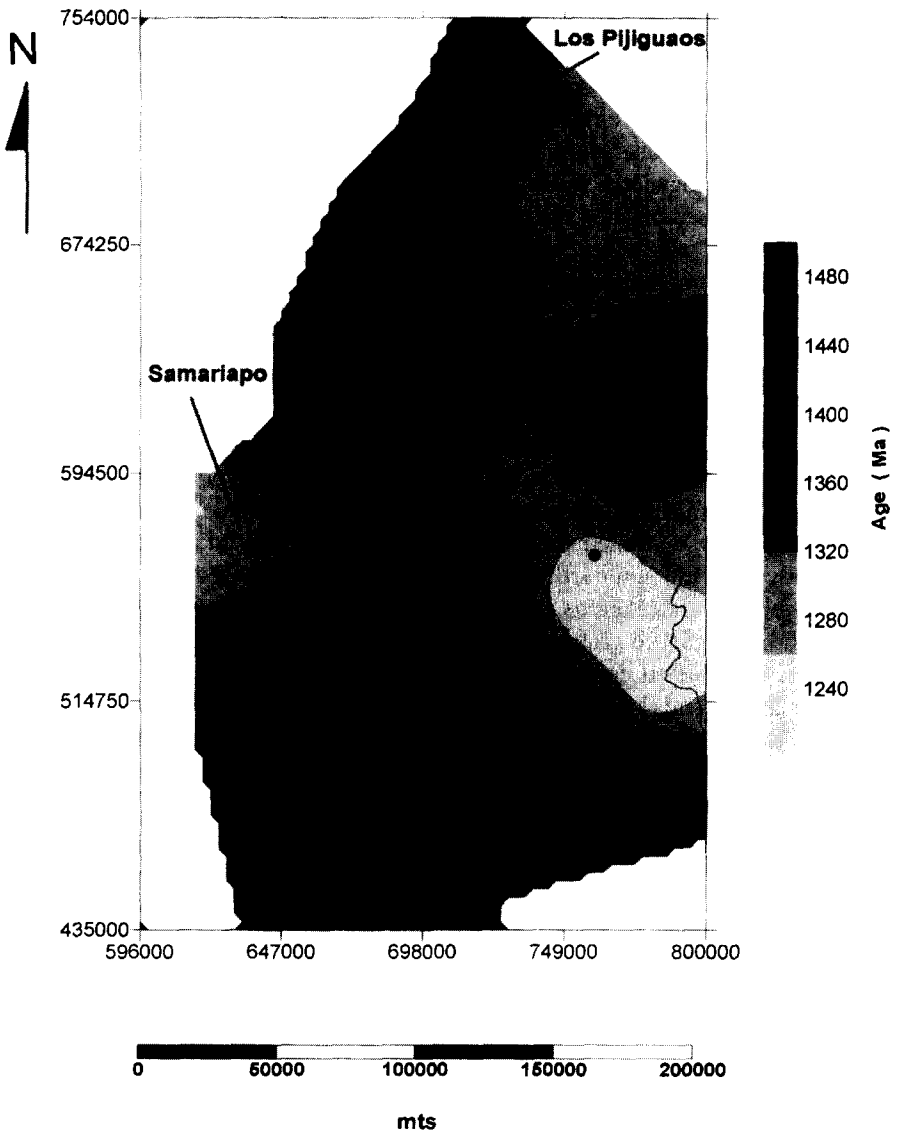


Fig. 15. Complete inferred "cooling map" of the Parguaza batholith and the PIC integrating Rb/Sr model ages (after Barrios, 1983) and paleomagnetic data (this study). A contour of the undifferentiated PIC (after Bellizia et al., 1976 and Wynn et al., 1993) is also shown along with the location of paleomagnetic (■) and geochronological sampling sites (●).

magnetic data with the information derived from Fig. 15. In fact, there are three different possible explanations for the spatial pattern of cooling observed in the Parguaza batholith and its regional extension down to the whole PIC:

1. By having, for different sites within the same batholith, only two mutually exclusive primary paleomagnetic components, together with manifold evidence for petrological continuity along the whole transect (Mendoza, 1974, 1976; Bangerter, 1985; Menéndez et al., 1985), it is reasonable to think of Parguaza as a single and slowly cooling intrusion.

Assuming that the Los Pijiguaos region is indeed the roof of Parguaza (Menéndez et al., 1985), and as far as one moves southwest along the transect approaches to the core of the batholith, it is possible to argue for an inverse pattern of cooling,

at least for this part of the intrusion. In fact, Fig. 15 shows, for the Parguaza batholith, a central area of older cooling ages flanked by a northeastern (site PI-1) and a southwestern (site CSP-2) region of lower cooling ages. This result agrees with the geochemical and petrogenetic studies by Mendoza et al. (1977) who, based on the variations of Mg/Fe ratios versus alkali percentage (Na_2O+K_2O), proposed a compositional inverse pattern of cooling for this pluton.

In addition, variations of Q_n values and the standard deviations of their site-averages along the transect (Fig. 7), seem to go against the most commonly known pattern of mineral grain sizes observed in normally zoned batholiths for which fast-cooling fine-grained minerals at their margins grade up to slow-cooling coarse-grained minerals at their centers. Moreover, if the standard deviations for the Q_n site-averages are taken as indicators of the range width of magnetite grain sizes in each of these sites, it is possible to conclude that such widths are larger at the ends of the transect than towards its middle part. In fact, it would be expected that those regions within the batholith which cooled first and faster had less time to develop a wide range of grain sizes, whereas those which cooled slower and last at the margins had more time to develop the range of grain sizes reflected by high values of the standard deviations of the site-averaged Q_n values.

Since it is difficult to envisage a physical process that could lead to a compositional and textural pattern of inverse cooling for a single intrusive body, we suggest two alternative hypothesis:

2. A sequential emplacement of magmas, controlled by normal faulting (continental rifting), could explain the regional cooling-age pattern observed in Fig. 15 for the PIC (including the Parguaza batholith). An important feature related to the long (ca 250 km) and wide (ca 100 km) first-cooled terrain, and its easterly lower-age counterpart (Fig. 15), are their northwesterly trends which coincide with the regional strike that typifies the Cuchivero Province in the Guyana Shield (Fig. 1). Thus, a causal relationship between the taphrogenic structural setting of the older units in this geological province (that probably resulted in its characteristic northwesterly grain, and the emplacement of the younger intrusions of the PIC) seems quite likely. Moreover, the two low-age areas that flank the high-age terrain, could be zones of high thermal gradient related to rifting at the sides of the first-cooled core of a major intrusive body, probably the down-sinking block proposed by Gaudette et al. (1978) to explain the origin of the Parguaza batholith and other intrusions in the Guyana Shield. The major problem with this theory is the difficulty of explaining how pure reworking of the continental crust, and consequent syn-rifting, could account for the generation of the large volumes of magmas that formed the PIC.

3. A third hypothesis, which also explains the apparently inverse zoning of the Parguaza batholith and its related anorogenic intrusions, has been recently proposed by Jácome Reyes et al. (1996a,b). This hypothesis entails the existence of a system of Precambrian and/or Triassic–Jurassic NE-trending horsts and grabens that cuts through the PIC intrusions. As a preliminary test, a gravimetric modeling has been carried out along a SW–NE, 420 km-long transect located at the northern lobe of the Parguaza batholith. According to this model, the already cooled and normally zoned Parguaza batholith has been faulted, vertically displaced and finally eroded in such a way that the surface reflection of its internal tectonism resulted in the apparent pattern of inverse zoning revealed by the map of Fig. 15. We favor this last hypothesis since it does not conflict with a physically feasible process of normal cooling for these intrusions, and with their anorogenic emplacement in a tectonic setting of subduction boundary (Gaudette and Olszewski, 1985; Gibbs and Barron, 1993). Such a structural context implies that there are viable mechanisms of the magma generation leading to the formation of large granitoid bodies. Moreover, the subsequent extensional regime that affected the Parguaza anorogenic intrusions, pivotal in Jácome Reyes et al. (1996a,b) interpretation, has been described at length by Gibbs and Barron (1993) as a series of events, of mainly extensional-vertical fault-block tectonics, that dominated the Shield from middle Proterozoic times to the present day. The most conspicuous of such events are the Nickerian stress

picture that occurred ca 1250 Ma ago, the development of the Takutu graben (Jurassic) in northeastern Brazil and close to the Venezuelan southeastern border, and the rifts that precede the opening of the Atlantic Ocean (Jurassic–Triassic).

7. Conclusions

Based on the interpretation of our paleomagnetic data, we can make the following reconstruction for the sequence of events that resulted in the acquisition of Parguaza's characteristic NRMs:

(1) By the time the different sites in the transect reached temperatures close to 580°C, the primary single-domain magnetites, titanomagnetites ($Fe_{2.7}Ti_{0.3}O_4$) and a small amount of deuteric hematite, recorded a magnetic remanence that, depending on the site, could be either G1R (CSP-3, PI-2 and PI-4) or G3N (CSP-2, PI-1 and PI-3).

(2) Some time after temperatures had dropped a few degrees below 580°C, there was an exsolution of submicron Ti-rich regions alternating with almost pure magnetite in multidomain titanomagnetite grains. This exsolution produced a second generation of single-domain-like magnetites (Ti-poor regions) with coercivities and unblocking temperatures spectra that overlap those of primary magnetic mineralogies. During exsolution, and for both G1 and G3, a reversal of the geomagnetic field took place, being recorded in these secondary magnetic carriers as either G1N or G3R. Hence the impossibility of separating, in 66% of the cases, the normal and reversed directions of remanence.

(3) There is neither geological nor petrological evidence for the alteration and/or metamorphism of the Parguaza rapakivi granites (Parguaza batholith). Therefore both G1 (N and R) and G3 (N and R) must be primary TRMs/TCRMs that record two different moments of the batholith's geological history.

(4) From the southwestern end of the transect, and according to the map of 'chrontours' (Fig. 3), it becomes clear that the G1 magnetization should be older than G3. A complete map of relative cooling ages for the Parguaza batholith can be constructed by extrapolating these results to the northeastern end of the transect where no radiometric data are available (Fig. 15). This map reveals a pattern of differential cooling for the Parguaza batholith with an older central region, and two younger margins at the northeastern and southwestern ends of the transect.

(5) Textural zoning along the transect is also revealed by the site-averages of the Q_n values and their corresponding standard deviations (Fig. 7). In fact, magnetite grain sizes seem to go from coarse-grained at the ends, down to fine-grained at the central part of our sampling profile.

(6) The pattern of differential cooling observed in the Parguaza batholith can be stretched south, to a regional scale, for the whole PIC (Fig. 15), being explained as the result of inverse zoning of a single intrusive body, however, this is difficult to envisage from a physical standpoint. Alternative hypothesis are, either a scenario of continental rifting, namely a sequential emplacement of magmas controlled by normal faulting (Gaudette et al., 1978), or the result of post-intrusion tectonism in the region rendering in a system of NE-trending horsts and grabens that cuts through the whole PIC and yields their actual apparent pattern of inverse zoning (Jácome Reyes et al., 1996a,b). We favor this scenario because it does not conflict with the plate tectonic context of a subduction boundary, acting in the western Venezuela Guyana Shield, and is assumed to explain the generation of the large volumes of magmas that gave rise to the Parguaza anorogenic intrusions.

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